Possible and Necessary Popularity in House Allocation PseudoCode

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1 Hypothesis

- Agents give their partial strict linear order (pairwise comparison).
- There are as many objects as agents.
- We consider only strict linear order.

2 Properties

Proposition 1 (Verify Possibly popular matching)**.** *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $\exists o \in O$ such that $o \succ_j \sigma(j)$ then return False (the matching is not popular).*

Proposition 2 (Find Possibly popular completion)**.** *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ for $o \in O$ such that $(\sigma(j), o) \notin P$ add $(\sigma(j), o) in \succ_j$*

Proposition 3 (Find Possibly popular completion - Rewritting)**.** *For a given possibly popular matching on a partial profile P, there is a complete completion that can be constructed as follows: If $\exists i, j \in N$ such that*

- *$\sigma(j)P_i\sigma(i)$ and for $o \in O$ such that $(\sigma(j), o) \notin P$ add $(\sigma(j), o) in \succ_j$*
- *$(\sigma(i), \sigma(j)) \notin P_i$, add $(\sigma(i), \sigma(j)) in \succ_i$*

Proposition 4 (Find Necessarily popular matching)**.** *If $\exists i, j \in N$ such that $\sigma(j)P_i\sigma(i)$ and $o \in O$*

- *$oP_j\sigma(j)$ or*
- *$(\sigma(j), o) \notin P_j$*

Return False (it is not a necessarily popular matching).

Proposition 5 (Find Necessarily popular completion)**.** *If $\exists i, j \in N$ such that $(\sigma(i), \sigma(j)), (\sigma(j), \sigma(i)) \notin P_i$ and $o \in O$ such that*

- *$oP_j\sigma(j)$*
- *$(\sigma(j), o) \notin P_j$*

Return False (it is not a necessarily popular matching).

3 Algorithms

3.1 Implementation with pairwise comparison

Algorithm 1 VerifPossiblyPopularMatching

Input: O the set of n objects, N the set of n agents, $P = (P_1, ..., P_n)$ the partial profile such that $P_i = [(o_l, o_k), ...]$ is the list of pairwise partial preferrence of agent i, $\mu = \{1 \colon \sigma(1), ..., n \colon \sigma(n)\}$ is the matching ($\sigma(i)$ is the assigned object to i in the matching).

Output: boolean indicating either the matching μ is possibly popular

for each agent $i \in N$ **do**

  for each preference $(o_{sup}, o_{inf}) \in P_i$ **do**

    NotFpost $\leftarrow$ o_{inf} is $\sigma(i)$

    NotDominatedByFpost $\leftarrow$ $False$

    for each preference $(o, o') \in P_j$ **do**

      NotDominatedByFpost $\leftarrow$ o_{sup} is o'

    end for

    if NotFpost and NotDominatedByFpost **then**

      return False

    end if

  end for

end for

return True

Algorithm 2 FindPossiblyPopularMatching

Input: O the set of n objects, N the set of n agents, $P = (P_1, ..., P_n)$ the partial profile such that $P_i = [(o_l, o_k), ...]$ is the list of pairwise partial preferrence of agent i, $\mu = \{1 \colon \sigma(1), ..., n \colon \sigma(n)\}$ is the perfect matching ($\sigma(i)$ is the assigned object to i in the matching).

Output: A complete completion that making the matching popular.

$\succ$ the complete completion $\leftarrow$ copy of the partial completion P

for each agent $i \in N$ **do**

  for each objects $sigma(j)$ and agents j matched $\in \mu$ **do**

    if $\exists(\sigma(j), \sigma(i)) \in P_i$ **then**

      for each objects $o \in O \backslash \{\sigma(j)\}$ **do**

        if $(\sigma(j), o) \notin P_j$ **then**

          Add $(\sigma(j), o)$ in $\succ_j$

        end if

      end for

    else if $\nexists(\sigma(i), \sigma(j)) \in P_i$ **then**

      Add $(\sigma(i), \sigma(j))$ in $\succ_i$

    end if

  end for

end for

return $\succ$

3.2 First implementation with positional ranking

Algorithm 3 VerifPossiblePopularMatching

Input: O the set of n objects, N the set of n agents, $P = (P_1, ..., P_n)$ the partial profile such that $P_i = [?, \sigma(i), o, ...]$ is the list of partial preferrence of agent i (? indicates unknown preference and $\sigma(i)$ is the assigned object to i in the matching), $\mu = \{1\colon \sigma(1), ..., n\colon \sigma(n)\}$ is the matching.
Output: boolean indicating either the matching μ is possibly popular, P a completion of the partial profile making μ possibly popular.

for each agent $i \in N$ **do**
  for each preference $p \in P_i$ **do**
    if p is unknown (equal to ?) **then**
      // Is there a completion such as p is the fpost matched to another agent in μ?
      if $\sigma(i)$ not in P_i **then**
        Assign $\sigma(i)$ to p
      else
        Find a matched fpost (should not be a fpost for more than half agents) and assigned it to p
      end if
    else if p is $\sigma(i)$ **then**
      Complete the preference list P_i with the remaining objects.
      Break
    else
      // p is an object with a better rank than $\sigma(i)$
      if p is a fpost matched to another agent **then**
        Continue
      else
        // p is not a matched fpost, the matching μ cannot be popular
        return False, P
      end if
    end if
  end for
end for
return result

Algorithm 4 VerifyNecessarilyPopularMatching

Input: O, N, $P = (P_1, ..., P_n)$, $\mu = \{(1, \sigma(1)), ..., (n, \sigma(n))\}$
Output: boolean indicating either the matching is necessarily popular or not.
$cpt_f post := 0$
for each $i \in N$ **do**
    $Isfpost := True$
        // A matching is popular if each agent obtained either its fpost or spost.
        // A fpost and/or a spost can not be specified by an agent at the beginning. So we check if all the possible complétion respect this condition.
    while There exist a completion of the agent's i preference list **do**
        $O_i :=$ the set of non ranked item by agent i.
        for Each unknown $p \in P_i$ **do**
            Choose an o in O_i for each unknown p
            $O_i \leftarrow O_i \ \{o\}$
        end for
            // At the end of this for loop, we get a completion of the agent's i preference list
            // Checking this completion respect the necessarily popular conditions
        if not $(\sigma(i) \in \{f(i), s(i)\})$ **then**
            return False;
        else if $\exists x \in P_i$ such that $x \succ_i \sigma(i)$ and x not a fpost **then**
            return False;
        end if
        if $\sigma(i) is not an f post in this completion$ **then**
            $Isfpost \leftarrow False$
        end if
    end while
    if $Isfpost$ **then**
        $cpt_f post \leftarrow cpt_f post + 1$
    end if
        // If at least half of fpost are assigned, the matching μ is necessarily popular
    if $cpt_f post \geq \lceil n/2 \rceil$ **then**
        return False;
    end if
end for
return True;

4 Examples

Instance for possibly popular matching Let H an instance of the House Allocation problem such as:

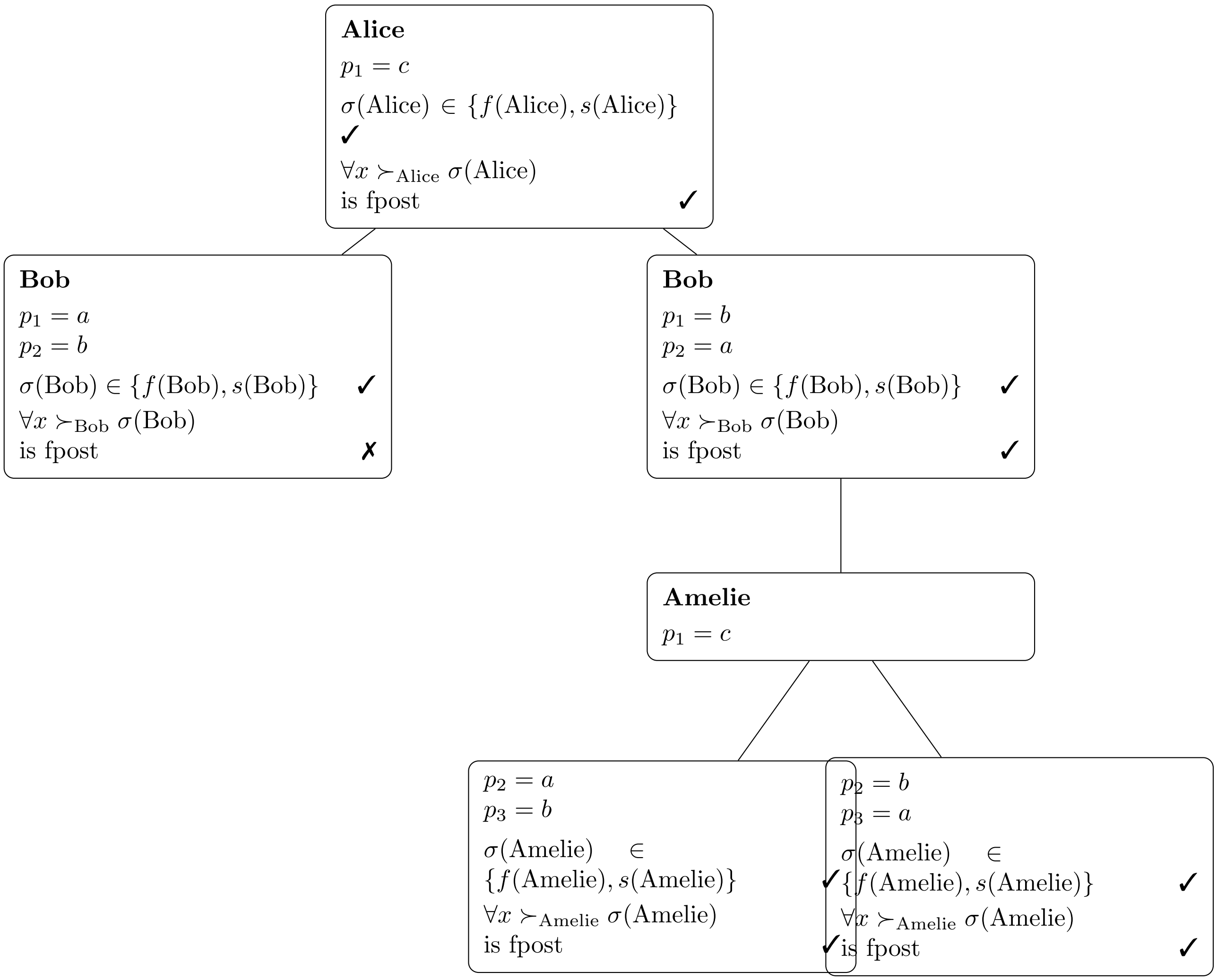
- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i.
- $\mu = \{(Alice, b), (Bob, c), (Amelie, a)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	**Completions**		
Alice	? a ?	$\boxed{b}$ a c	$\boxed{b}$ a c	c a $\boxed{b}$
Bob	b ? ?	b $\boxed{c}$ a	b $\boxed{c}$ a	b $\boxed{c}$ a
Amelie	? a ?	c $\boxed{a}$ b	b $\boxed{a}$ a	b $\boxed{a}$ c
		not popular	no solution	not popular

Instance for necessarily popular matching - Non necessarily popular matching Let H an instance of the House Allocation problem such as:

- $N = \{Alice, Bob, Amelie\}$ the set of agents
- $O = \{a, b, c\}$ the set of objects (houses)
- $P = \{P_1, P_2, P_3\}$ the partial profile for which P_i is the strict partial preference order of the agent i.
- $\mu = \{(Alice, a), (Bob, b), (Amelie, c)\}$ the matching. Some object specified in the matching are not ranked in some preference lists.

Agents	P_i	**Completions**		
Alice	? a b	$\boxed{b}$ a c	$\boxed{b}$ a c	c a $\boxed{b}$
Bob	? ? a	b $\boxed{c}$ a	b $\boxed{c}$ a	b $\boxed{c}$ a
Amelie	c ? ?	c $\boxed{a}$ b	b $\boxed{a}$ a	b $\boxed{a}$ c
		not popular	no solution	not popular


Alice
$p_1 = c$
$\sigma(\text{Alice}) \in \{f(\text{Alice}), s(\text{Alice})\}$ ✓
$\forall x \succ_{\text{Alice}} \sigma(\text{Alice})$ is fpost ✓
Bob
$p_1 = a$
$p_2 = b$
$\sigma(\text{Bob}) \in \{f(\text{Bob}), s(\text{Bob})\}$ ✓
$\forall x \succ_{\text{Bob}} \sigma(\text{Bob})$ is fpost ✗
Bob
$p_1 = b$
$p_2 = a$
$\sigma(\text{Bob}) \in \{f(\text{Bob}), s(\text{Bob})\}$ ✓
$\forall x \succ_{\text{Bob}} \sigma(\text{Bob})$ is fpost ✓
Amelie
$p_1 = c$
$p_2 = a$
$p_3 = b$
$\sigma(\text{Amelie}) \in \{f(\text{Amelie}), s(\text{Amelie})\}$ ✓
$\forall x \succ_{\text{Amelie}} \sigma(\text{Amelie})$ is fpost ✓
$p_2 = b$
$p_3 = a$
$\sigma(\text{Amelie}) \in \{f(\text{Amelie}), s(\text{Amelie})\}$ ✓
$\forall x \succ_{\text{Amelie}} \sigma(\text{Amelie})$ is fpost ✓

5 Proof

5.1 Proof Verif Possible Popular Matching

Hypotheses

- N is a set of n agents and O a set of n objects.
- We assume a **perfect matching**: $|N| = |O|$ and all agents/objects are matched.
- Preferences are given as **partial orders** (pairwise comparisons).
- We consider **completions of the partial preference profile**.
- μ is a matching.

Definitions

- A **completion** is a strict complete order extending the partial preferences.
- A matching μ is **popular** if for every matching μ':

$$|\{i \in N : \mu(i) \succ_i \mu'(i)\}| \geq |\{i \in N : \mu'(i) \succ_i \mu(i)\}|.$$

- μ is **possibly popular** if it is popular under at least one completion of the partial preference profile.

Proof of $\Rightarrow$ Suppose that there exist $i, j \in N$ such that

$$\sigma(j) P_i \sigma(i) \quad \text{and} \quad \exists o \in O \text{ such that } o \succ_j \sigma(j).$$

We prove that μ is not possibly popular.

Let $\succ$ be any completion of the partial preference profile P. We show that μ is not popular under $\succ$ by constructing a matching μ' that defeats μ.

Construction of μ'

- If o is unmatched in μ, define:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o.$$

- Otherwise, there exists $k \in N$ such that $o = \sigma(k)$. Define a local exchange:

$$\mu'(i) = \sigma(j), \quad \mu'(j) = o, \quad \mu'(k) = \sigma(i),$$

and for all $\ell \notin \{i, j, k\}$:

$$\mu'(\ell) = \mu(\ell).$$

Validity

The matching μ' is valid since it corresponds to a permutation of objects among at most three agents and leaves all others unchanged.

Comparison of μ and μ'

- Agent i prefers μ':

$$\sigma(j) \succ_i \sigma(i).$$

- Agent j prefers μ':

$$o \succ_j \sigma(j).$$

- At most one agent (namely k if it exists) may prefer μ.
- All other agents are indifferent.

Vote count

$$|\{i \in N : \mu'(i) \succ_i \mu(i)\}| \geq 2 \quad \text{and} \quad |\{i \in N : \mu(i) \succ_i \mu'(i)\}| \leq 1.$$

Thus, μ' is more popular than μ, contradicting the popularity of μ.

Since this holds for any completion, μ is not possibly popular.

□

Proof of $\Leftarrow$ Suppose that μ is not possibly popular. Then for every completion of the partial preference profile, μ is not popular.

We use the following characterization from Abraham et al.:

Lemma (Abraham et al.) A matching μ is popular if and only if:

- every agent is matched to either its *fpost* or its *spost*,
- and every *fpost* is matched.

By contraposition of the lemma (Abraham et al), since μ is not popular for any completion, for every completion there exists at least one agent i such that:

$$\mu(i) \notin \{fpost(i), spost(i)\}.$$

In particular, this implies:

$$fpost(i) \succ_i \mu(i).$$

Let $j \in N$ be such that:

$$\mu(j) = fpost(i).$$

Thus:

$$\sigma(j) \succ_i \sigma(i).$$

Let $k \in N$ be such that:

$$\mu(k) = spost(i).$$

We distinguish the following cases:

1. If $\mu(j) \notin \{fpost(j), spost(j)\}$, then by definition of $fpost(j)$ there exists an object $o \in O$ such that:

   $$o \succ_j \mu(j),$$

   and the property holds.

2. If $\mu(j) = spost(j)$, then by definition of $fpost(j)$ we have:

   $$fpost(j) \succ_j \mu(j),$$

   so there exists $o = fpost(j)$ such that:

   $$o \succ_j \sigma(j).$$

3. If $\sigma(j) = fpost(j)$ and $\sigma(k) = fpost(k)$, then both agents receive their *fpost*.

   Since μ is assumed not popular, this contradicts the characterization of Abraham et al., which requires that every agent is matched to either its *fpost* or *spost*, and that every *fpost* is matched. Hence, this situation cannot occur under our assumption.

In all cases, we obtain:

$$\exists o \in O \text{ such that } o \succ_j \sigma(j).$$

Therefore:

$$\exists i, j \in N \text{ such that } \sigma(j) \succ_i \sigma(i) \quad \text{and} \quad \exists o \in O \text{ such that } o \succ_j \sigma(j).$$

This proves the desired property.

□

5.2 Proof Find Possibly Popular Completion

Hypotheses

- N is a set of n agents and O a set of n objects.
- We assume a **perfect matching**: $|N| = |O|$ and all agents/objects are matched.
- Preferences are given as **partial orders** (pairwise comparisons).
- We consider **completions of the partial preference profile** P $\succ$.
- μ is a possibly popular matching.

Definitions

- $\forall i \in N, \forall o \in O \backslash \{\sigma(i)\}$, $\exists (\sigma(i), o)$ or $(o, \sigma(i))$ in $\succ_i$.
- **From Handbook of Computational Social Choice (p26):** $a \gtrsim b$ or $b \gtrsim a$ for all $a \neq b \in O$ and any voter $i \in N$

Proof We suppose μ is a possibly popular and perfect matching. Suppose we have found a completion that makes μ popular such that $\exists i, j \in N$:

- if $\sigma(j) P_i \sigma(i)$ and for $o \in O$ such that $(\sigma(j), o) \notin P_j$ then $(\sigma(j), o) \in \succ_j$
- if $(\sigma(i), \sigma(j)) \notin P_i$ then $(\sigma(i), \sigma(j)) \in \succ_i$

Let's prove that this completion is indeed complete and that μ is a popular matching under this completion.
Intuition: O is the set of all objects and because μ is a perfect matching, O contains all the matched objects. So we can rewritte O as $O = \{\sigma(k) : k \in N\}$. For each object $\sigma(k) \in O \backslash \{\sigma(i)\}$ and for each agent $i \in N$,

- either $(\sigma(i), \sigma(k)) \in \succ_i$
- or $(\sigma(k), \sigma(i))$ and $(\sigma(k), o) \in \succ_j$ for all $o \in O$

From the definition of a possibly popular matching μ, we know that there exists a completion $\succ$ of the partial preference profile P such that μ is popular under $\succ$. Under this completion and from the Abraham et al. characterization of popular matchings, we have that every agent is matched to either its *fpost* or its *spost*, and every *fpost* is matched. Hence:

1. Because there are as many agents as houses, for all $i \in N$, $\mu(i)$ is a *fpost* of any agent j such that $fpost(j) \succ_i \mu(i)$.
2. For all $\mu(i)$, if there does not exist an agent j such that $\mu(i) = fpost(j)$, then μ is not popular under $\succ$, which is a contradiction of the second condition of the characterization of Abraham et al.
3. For all $\mu(i)$, if $\mu(i) \succeq_i spost(i)$ then the first condition of the characterization of Abraham et al. is verified. In the other case, when $spost(i) \succ_i \mu(i)$, the first condition is not verified and μ is not popular under $\succ$.